

Ligjerata 11dh

Ekuacionet i dytë Puasonit

$$pV^k = \text{konst} \quad \text{E. I. Puasonit}$$

$$pV = RT \Rightarrow p = \frac{RT}{V}$$

$$\frac{RT}{V} \cdot V^k = \text{konst.}$$

$$T \cdot V^{k-1} = \text{konst.}$$

$$T \cdot V^{k-1} = \text{konst.} \quad \underline{\text{E. II. Puasonit.}}$$

shkurte E.I. P.

$$dQ = du + dA \quad \text{pra } dA = pdV$$

$$dQ = du + pdV$$

$$dQ = 0$$

$$du = -pdV, \quad du = C_v dT$$

$$pV = nRT, \quad n = 1 \text{ mol}$$

$$pV = RT$$

$$p = \frac{RT}{V}$$

$$C_p = C_v + R$$

$$n = 1 \text{ mol}$$

$$du + pdV = 0$$

$$R = C_p - C_v$$

$$C_v dT + \frac{RT}{V} dV = 0$$

$$C_v dT + \frac{C_p - C_v}{V} T dV = 0 \quad / : C_v \cdot T$$

$$\frac{dT}{T} + \frac{C_p - C_v}{C_v} \cdot \frac{dV}{V} = 0$$

$$\frac{dT}{T} + \left(\frac{C_p}{C_v} - 1 \right) \cdot \frac{dV}{V} = 0$$

$$\frac{C_p}{C_v} = k$$

$$\frac{dT}{T} + (k - 1) \frac{dV}{V} = 0 \quad / \int$$

$$\int_{T_1}^{T_2} \frac{dT}{T} + (k - 1) \int_{V_1}^{V_2} \frac{dV}{V} = 0$$

ose

$$(k - 1) \int_{V_1}^{V_2} \frac{dV}{V} = - \int_{T_1}^{T_2} \frac{dT}{T}$$

$$(k - 1) (\ln V_2 - \ln V_1) = - (\ln T_2 - \ln T_1)$$

$$(k - 1) (\ln V_2 - \ln V_1) = (\ln T_1 - \ln T_2)$$

$$(k - 1) \ln \frac{V_2}{V_1} = \ln \frac{T_1}{T_2}$$

$$(k - 1) \ln \frac{V_2}{V_1} = \ln \frac{T_1}{T_2}$$

$$\left(\frac{V_2}{V_1} \right)^{k-1} = \frac{T_1}{T_2}$$

$$\boxed{T_1 V_1^{k-1} = T_2 V_2^{k-1}} \quad *$$

$$\boxed{T \cdot V^{k-1} = \text{konst.}}$$

Ekuacionet i tretë Puasonit

Nga e I i Puasonit

$$p \cdot V^k = \text{konst.}$$

$$p_1 V_1^k = p_2 V_2^k$$

$$\left(\frac{V_1}{V_2} \right)^k = \frac{p_2}{p_1} \quad \text{ose} \quad \frac{V_1}{V_2} = \left(\frac{p_2}{p_1} \right)^{\frac{1}{k}}$$

$$p_1 V_1 = RT_1 \Rightarrow V_1 = \frac{RT_1}{p_1}$$

$$p_2 V_2 = RT_2 \Rightarrow V_2 = \frac{RT_2}{p_2}$$

$$\frac{\frac{RT_1}{p_1}}{\frac{RT_2}{p_2}} = \left(\frac{p_2}{p_1} \right)^{\frac{1}{k}} \Rightarrow \frac{p_2 T_1}{p_1 T_2} = \left(\frac{p_2}{p_1} \right)^{\frac{1}{k}}$$

$$\boxed{\left(\frac{p_2}{p_1} \right)^{\frac{1-k}{k}} = \left(\frac{p_1}{p_2} \right)^{\left(\frac{1-k}{k} \right)}}$$

$$\frac{T_1}{T_2} = \frac{p_1}{p_2} \left(\frac{p_2}{p_1} \right)^{\frac{1}{k}} = \left(\frac{p_2}{p_1} \right)^{\frac{1}{k} - 1} = \left(\frac{p_2}{p_1} \right)^{\frac{1-k}{k}}$$

$$\frac{T_1}{T_2} = \left(\frac{p_1}{p_2} \right)^{\frac{k-1}{k}}$$

$$\left(\frac{T_1}{T_2} \right)^k = \left(\frac{p_1}{p_2} \right)^{k-1} \Rightarrow \frac{T_1^k}{T_2^k} = \frac{p_1^{k-1}}{p_2^{k-1}}$$

$$T_1^k \cdot T_2^{-k} = p_1^{k-1} \cdot p_2^{-(k-1)}$$

$$T_1^k \cdot \frac{1}{p_2^{k-1}} = \frac{1}{T_2^k} \cdot p_2^{-(k-1)}$$

$$T_1^k \cdot p_1^{1-k} = T_2^k \cdot p_2^{1-k}$$

$$T^k \cdot p^{1-k} = \text{konst} \quad \text{E. III i Puasonit}$$

$$a^2 = b / \sqrt{}$$

$$a = \sqrt[3]{b} = b^{1/3}$$